

Micro Syllabus

Subject: Data Communication II/II

Marks Allotment	Theory	Practical	Total
Sessional	40		40
Final	60		60
Total	100	25	125

Time Allotment	Lecture	Tutorial	Practical
Weekly Hours			
Depth Codes	C – Circuit	D – Definition	Dm – Demonstration
Dr – Derivation	Dw – Drawing	E – Explanation	I – Illustration
N – Numerical	P – Proof	Pr – Programming	S – State

Unit	Topic/Subtopic	Depth Code	Description of Depth	Actual Plan
				Hours
1	Introduction			6
	1.1 Data communication, data representation, Data Types, and flows	T	Key Terminologies: Data, Message, Information Signals, Communication Numbers, text, images, videos Types: Analog and Digital communication model, their advantages and disadvantages Transmission modes: Simplex, half-duplex, duplex	1.5
	1.2 Evolution of data communication	D, E, I	Briefly explain the history of data communication.	1
	1.3 Communication model	D, E, I	Basic and generic block diagrams in different modes	1.5
	1.4 Networks(LAN, MAN, WAN), Simplified network architecture: OSI model, TCP/UDP	D, E, I	LAN, MAN, WAN Function of the 7 layers of the OSI model in brief	1
	1.5 Data communication and networking for today's enterprise	E	Briefly explain Connectionless and connection-oriented protocols Applications in various wired and wireless communication frameworks, basics of IoT, and data computing models.	1

2	Data Communication Fundamentals: Signals and Systems			9 hours
	2.1 Analog and Digital Signals	D,E	Definition and types: basic and composite signals, dimensions: single, multi Channel: single, multi	1

	2.2 Analog and Digital signals: Continuous and Discrete time signals, 2.2.1 Classification of Signals: i) Periodic and Non-periodic Signals ii) Even and Odd signals iii) Deterministic and Non-Deterministic Signals iv) Energy and Power Signals, Continuous Time and Discrete Time Signals v) Causal, Non-causal and Anti-causal signals	D, E, I, Numerical	Graphically represent the signals Numerical Example: Energy and power, Test for Periodicity, causal and anti-causal, Even and odd part	2.5
	2.3 Systems 2.3.1 Basic Properties 2.3.2 Classifications: i) Static and Dynamic ii) Linear and non-linear iii) Time variant and invariant iv) Causal and non-causal v) Stable and unstable vi) Feedback and non-feedback	D, E, I, N	Test for linearity, Time invariance, and causality For others: Explain with examples only	2.5
	2.4 Fourier series and Fourier Transform 2.4.1 Conditions, properties, Frequency domain representation (magnitude and phase plot)	D, Dr E, I, N	Conditions, properties, Frequency domain representation (magnitude and phase plot)	2
	2.5 Transmission Impairments: Attenuation, distortion, noise	D, E, I	Explain each impairment with suitable diagrams	0.5
	2.6 Channel capacity: Nyquist bandwidth, Shannon-Hartley capacity Theorem	D, E, N	Formula and numerical	0.5
	2.7 Performance parameters	D, E	Briefly explain Bandwidth, Channel capacity, SNR, throughput, latency, jitter, bit rate, and baud rate	0.5

3	Transmission media and Data compression			8
	3.1 Guided transmission media: Co-axial cable, twisted pair, optical fiber	D, E, I	Briefly explain the transmission media (Guided), their maximum bandwidth and data rate, advantages, and disadvantages	1

	3.2 Unguided transmission media: Radio waves, microwaves, infrared	D, E, I	Briefly explain the transmission media (Un-Guided), their frequency range, and data rate, advantages, and disadvantages	1
	3.3 Wireless propagation (Introduction to groundwave propagation, sky wave propagation, and line of sight propagation), frequency bands	D, E, I	Describe the propagation modes with a suitable diagram. Mention the frequency for each propagation mode	1
	3.4 Error detection and correction: Error Detection: Parity, Check Sum, Cyclic Redundancy Check	D, E, I, N	Numerical examples for error detection methods	1
	3.5 Single Bit Error Correction: Hamming Code, Systematic Linear Block Coding/Decoding Codes (LBC) Codes	D, E, I, N	Numerical examples for error correction methods	2
	3.6 Data Compression: Entropy, Transmission Efficiency Lossless Compression Huffman Coding/Decoding Shannon-Fano Coding/Decoding	D, E, N	Numerical examples for Lossless data compression methods	2

4	Signal Encoding Techniques			12
	4.1 Performance Parameters: Synchronization, Bandwidth Required	D, E	Briefly explain the performance parameters for Analog-Analog, Analog-Digital, Digital-to-Analog, and Digital-to-Digital signal encoding Techniques	1.5
	4.2 Analog-to-Analog(Analog data, Analog Signal): Modulation and its need, AM, FM, PM	D, E, Dr, I, N	Define modulation and explain why we need modulation Graphically represent AM, FM, and PM Derive the standard equation of AM and FM Numericals from AM and FM [modulation index, bandwidth, Power delivered] Single-tone analysis of AM and FM Total Average power delivered in AM and PM Mention the Efficiency of AM Narrowband, Wideband FM formula	3

	4.3 Analog-to-Digital (Digital data, Analog signals): PAM, PWM, PPM, PCM, DM	D, E, I, Numerical	Figure waveforms Only impart knowledge of the Generation of PAM, PWM, PPM, PCM, DM using the respective Modulator block diagram Data rate calculation in PCM	2.5
	4.4 Digital-to-Analog (Digital data, Analog signal): ASK, FSK, PSK, QPSK, QAM	D, E, I	Only impart knowledge of the Generation of ASK, FSK, PSK, QPSK, QAM using respective modulator block diagrams Constellation diagram of ASK, FSK, PSK, QPSK, 8-QAM, 16-QAM, 64-QAM Eye diagram	3
	4.4 Digital-to-Digital (Digital data, Digital signal)	D, E, I	Define Line coding and the various techniques used in line coding Example waveforms for: Unipolar: NRZ, RZ Polar: NRZ-L, NRZ-I, RZ Bipolar AMI Scrambling techniques: B8ZS, HDB3	2

5	Multiplexing and Switching			6
	5.1 Introduction to Multiplexing and Multiple Access	D, E	Briefly explain bandwidth utilization techniques Define multiplexing and its need Define multiple access Difference between multiplexing and multiple access Application of multiplexing	0.5
	5.2 Frequency division Multiple access	D, E, I	Explain the working of the FDMA using a suitable Diagram, also include how a band-pass filter is used on the receiver side in FDMA Briefly describe the FDM hierarchy with a suitable block diagram	1
	5.3 Time division Multiple access	D, E, I	Explain the working of TDMA using a suitable Diagram Describe Synchronous TDMA and Statistical TDMA along with their advantages and Disadvantages	1

	5.4 Subscriber channel Division: T1 and E1 hierarchies	D, E, I	Mention that T1 and E1 are TDM hierarchies Number of channels used in T1 and E1 Hierarchies, with each channel's capacity Data rate supported by T1 and E1 hierarchies	0.5
	5.5 Spread spectrum:DHSS, FHSS, CDMA	D, E, I	What is spread spectrum with a simple block diagram Explain the working of DHSS with its block diagram. What is a chip sequence? Explain the working of FHSS with its block diagram. What is a pseudorandom code? What is CDMA? Near-far problem in CDMA Example diagram of 4 stations communicating with each other using CDMA	1.5
	5.6 Circuit Switching Network	D, E, I	What is a switch? Concepts of circuit switching and message switching with a suitable diagram	0.5
	5.7 Packet Switching Network	D, E, I	Briefly explain IP addresses and their use for routing. What is Packet Switching, with a diagram? Virtual switching Datagram switching Difference between circuit and packet switching.	1

6	Cellular Wireless Communication and Latest Trends			4
	6.1 Overview of 1G, 2G, 3 G,4G	D, E	Briefly review wireless communication. Mention the Timeline, Technology, Data Speed, Key Features, Limitations of each generation (G) Mention about VOLTE.	1
	6.2 Cellular Technology, Fundamental terminology: Cell, Frequency reuse, cluster, adjacent cell interference, co-channel interference, handoff-strategies.	D, E, I	Briefly explain all terminologies with suitable diagrams.	2

	6.3 Introduction to 5G networks, IOT communication	D, E, I	What is 5G? Features of 5G (speed, latency) Technology behind 5G(mmWave, MIMO, Beamforming) What is IOT? Application of 5G in IOT with a suitable diagram showing a network of nodes.	1
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Tutorials

1. Surveys the specification:
 - a. Protocols of 1G to 5G.
 - b. Communication channel categories of each guided medium.
2. Examples and proofs of different signal and system types.
3. Line coding: Examples and waveform generations.
4. Channel Coding: Waveform generations, encoding, decoding.

Practical

(22.5 hours)

1. Signal analysis using MATLAB
2. Implementation of small network using hub and switch in physical or simulation environment
3. Analog modulation generation and reconstruction
4. Pulse modulation generation and reconstruction
5. Conversion of given binary sequence into different line coding
6. Digital modulation (ASK, FSK, PSK) generation and reconstruction

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below.

Chapter	Hours	Marks Distribution*
1	6	6
2	9	12
3	8	11
4	12	16
5	6	10
6	4	5
Total	45	60

* There may be a minor deviation in the marks distribution.

Reference

1. Stallings, W. (2007). Data and Computer Communications. India: Pearson Education.
2. Forouzan, B. A., Fegan, S. C. (2007). Data Communications and Networking (McGraw-Hill Forouzan Networking). United Kingdom: McGraw-Hill Higher Education.
3. Tanenbaum, A. S., Wetherall, D. (2011). Computer Networks. India: Pearson Prentice Hall.
4. Rappaport, T. S. (2024). Wireless Communications: Principles and Practice. United Kingdom: Cambridge University Press.